Fair vehicle routing with random route durations: ex-post ordered fairness, rank prices and risk-consistent tours

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Abstract

Fair vehicle routing minimises an ordered measure of route lengths, such as the range or the Gini mean difference, and recent exact approaches require every route to be a shortest tour of its customers. They assume deterministic travel times, whereas drivers experience realised durations. We study fair routing with random travel times and distinguish ex-ante fairness, of expected workloads, from ex-post fairness, the expected unfairness of the realised day. We prove that deterministic fair plans overstate ex-post fairness, bound the gap, and show that it is largest for balanced plans. We give an exact, sampling-free integral for the expected value of any ordered measure of independent route durations, based on the Poisson–binomial distribution, and derive rank prices, the derivatives of expected unfairness with respect to each route. Because even the minimax objective can prefer a longer tour under uncertainty, we define route consistency through each driver's mean and standard deviation and show that the consistent tours are supported Pareto tours. A bilevel local search uses the rank prices to screen moves. In seventeen experiments on two families of benchmark instances, deterministic fair plans understate realised unfairness by 8 to 40 percent. Optimising ex-post fairness improves realised fairness by about 1 percent at the standard budget, 2 percent on a second instance family (better on all twelve instances) and up to 7 percent with generous budgets, while using less of the budget. It outperforms sample average approximation and a risk-adjusted baseline that is no better than ignoring uncertainty.

Keywords: Routing; Fairness; Stochastic travel times; Order statistics; Bilevel optimization

1 Introduction

Routing plans that minimise total distance often give some drivers much longer days than others. Unequal workloads hurt driver satisfaction and retention and can be seen as unfair (Matl, Hartl, & Vidal, 2018). The fairness literature in routing therefore optimises a *fairness measure* of the vector of route workloads: the longest route, the range, the Gini mean difference (GMD), the variance or a mean absolute deviation (Bektaş & Letchford, 2020; Karsu & Morton, 2015; Matl et al., 2018). Ordered measures, which weight the sorted workloads (Nickel & Puerto, 2005; Ogryczak & Śliwiński, 2003; Yager, 1988), cover all of these in one framework (Blanco, Pozo, Puerto, & Torrejon, 2026).

Non-monotone measures such as the range or the GMD create a well-known pathology: an optimiser can lower the measure by making a light route *longer*, giving a driver a needless detour (Matl et al., 2018). van Rossum, Chen, and Lodi (2026a, 2026b) and Ljubić, Puerto, and Torrejón (2026) close this loophole exactly, by requiring every route to be a shortest tour of its customers, i.e. an optimal solution of the travelling salesman problem (TSP) on its customers. Ljubić et al. (2026) write the problem as a bilevel programme in which a planner assigns customers and each driver solves a TSP, and solve instances with up to 35 customers to optimality for seven measures.

All of these works assume deterministic travel times. A driver's day, however, is a realised duration: congestion, weather, parking and service-time variability differ between routes and between days. Fairness is then a statement about random variables, and three questions arise that the deterministic theory does not ask.

1. *What should be balanced?* The expected workloads (ex-ante fairness), or the workloads drivers actually experience, sorted day by day (ex-post fairness)? The identity of the worst-off driver changes from day to day, and the two notions can disagree.

2. *How can ex-post fairness be optimised?* Sample-average approximation (SAA) is the default, but sorting per scenario makes the objective noisy and its gradient uninformative.

3. *What is a consistent route?* Under random durations "the shortest tour" is not obviously what a driver would choose, and we show that monotone measures no longer protect drivers from detours.

Contributions.

- **Model.** We formulate fair routing with random travel times (a fleet-wide day factor times arc-level noise) as a bilevel problem with an ex-post ordered fairness objective, an expected-cost budget and per-driver risk-consistent tours. With no noise it reduces to the fair capacitated vehicle routing problem with shortest-tour routes (X-CVRP-TSP) of Ljubić et al. (2026).

- **Fairness illusion (Proposition 9).** For every ordered measure with nondecreasing weights, ex-ante fairness underestimates ex-post unfairness, by at most $\|\lambda\|_1 v(\sum_k W_k)^{1/2}$, where W_k is the route's quadratic length. Common multiplicative shocks leave the expected value of every ordered measure unchanged (Proposition 3); they do inflate the variance measure and the tail of unfairness.

- **Poisson–binomial order-statistic integral (PBOI; Theorem 5).** An exact, sampling-free formula for the expected ordered measure of independent route durations with given marginals (we use moment-matched gamma marginals), as an integral of a Poisson–binomial distribution against tail sums of the weights. It handles negative weights and costs $O(Gm^2)$.

- **Rank prices (Proposition 7).** The derivative of expected unfairness with respect to a location shift of a route's duration equals the expected weight of its rank. Prices of dispersion measures sum to zero, and a negative price is precisely the stochastic incentive to lengthen a route.

- **Risk-consistent tours (Proposition 11, Theorem 13).** Even $\mathbb{E}\max$ can prefer a non-shortest tour. We define consistency through the driver's own mean–standard deviation and prove that the consistent tour is a supported Pareto tour of (length, quadratic length), found exactly by dichotomic parametric Held–Karp.

- **Algorithm and evidence.** SORTIE (Stochastic Ordered Route fairness with Tour-consistency via Integral Evaluation), a bilevel local search whose moves are screened by rank prices, is compared with the deterministic fair plan, SAA with up to 5000 scenarios, a risk-adjusted deterministic baseline and ablations, in seventeen experiments: on the instances used by the deterministic literature and on a second family of twelve CVRPLIB

instances, with exact enumeration on small instances, budget, fleet-size, noise-level and risk-attitude sweeps, four noise models including correlated noise, and up to 100 customers.

2 Literature review

SORTIE sits at the intersection of four literatures: workload equity in deterministic vehicle routing, inequity-averse optimisation with ordered measures, fairness under uncertainty, and routing with stochastic travel times. We review each in turn, stating for the closest works what they model, how they solve it and what they leave open, and close with the tools we borrow from probability and multicriteria optimisation. Table 1 summarises the positioning.

2.1 Workload equity in deterministic routing

The systematic study of balanced routing starts with Matl et al. (2018), who compare the equity functions used in the literature (min-max, range, mean absolute deviation, Gini-type measures and others) on the same instances. Their central observation is that the measures differ in *monotonicity*: a measure that is not monotone in the individual workloads can be improved by making a short route longer, so an optimal "fair" plan may contain routes that are needlessly inefficient. Matl, Hartl, and Vidal (2019b) show that the choice of the workload resource (distance, duration, load or number of customers) changes the balanced plan and the cost of equity, and Matl, Hartl, and Vidal (2019a) approximate the cost–balance Pareto front with a box-splitting scheme around a single-objective heuristic. In all three works the workload is a deterministic quantity attached to a route.

Exact methods have followed two directions. The min-max (makespan) objective has a long tradition (Applegate, Cook, Dash, & Rohe, 2002), and Bertazzi, Golden, and Wang (2015) quantify how much longer the longest route can become when total cost is minimised instead. For balance objectives, Halvorsen-Weare and Savelsbergh (2016) study bi-objective general routing with several balance criteria, Bektaş and Letchford (2020) propose ℓ_p norms of the workload vector as a tunable family between efficiency and min-max fairness, and Agius, Absi, Feillet, and Garaix (2026) develop exact algorithms for routing with route-cost equity *constraints*. van Rossum, Chen, and Lodi (2024); van Rossum et al. (2026a) design branching rules for range and order-based objectives that make the range tractable in branch-and-price, and van Rossum et al. (2026b) add cutting planes that forbid arc sequences dominated by a shorter tour of the same customers, which removes exactly the inefficiency identified by Matl et al. (2018). Related models balance tours in the multiple travelling salesman problem with a bound on the Gini coefficient of tour lengths (Bhadoriya, Deka, & Sundar, 2026), trade egalitarian against utilitarian welfare in cooperative fleets (López Sánchez, Lujak, Semet, & Billhardt, 2024), weigh the cost of driver satisfaction (Bruinink, Berghman, & Dollevoet, 2025), equalise *cumulative* workload across periods by assigning daily routes to drivers (Nekooghadirli, Gendreau, Potvin, & Vidal, 2026), and allocate workload among heterogeneous drivers by Nash social welfare under a cost cap (Mahmoodian, Charkhgard, & Dayarian, 2025), and design equity-oriented workload and pay for crowdsourced drivers (Sobhanan, Charkhgard, & Dayarian, 2026).

The direct predecessor of this paper is Ljubić et al. (2026). They write a whole family

of fairness objectives (maximum, range, Gini mean difference, absolute deviation, variance) as ordered measures of the vector of route lengths, borrowing the ordered-median machinery of location theory, and resolve the monotonicity problem of Matl et al. (2018) with a *bilevel* formulation: the upper level chooses the partition of the customers, and each lower level must route its customers on a shortest tour. They solve the resulting X-CVRP-TSP exactly on instances sliced from CVRPLIB X-n641-k35, which we reuse. Their model, like every work in this subsection, is deterministic; their conclusion lists uncertain demands as future work and does not discuss travel-time uncertainty. SORTIE keeps their bilevel structure and their ordered measures but lets route durations be random. Three things change. The objective becomes an expectation of an ordered measure of random variables, which no deterministic formulation represents (Section 4). Monotonicity no longer guarantees that the optimal tours are shortest ones (Proposition 11), so route consistency must be imposed for every measure, not only for the non-monotone ones. And "shortest" itself needs a definition when the duration of a tour is a random variable; we give one through each driver's mean–standard-deviation criterion and show that it is computable (Theorem 13).

2.2 Ordered measures and inequity-averse optimisation

Ordered weighted averaging was introduced by Yager (1988) as an aggregation that weights outcomes by their rank rather than by their identity; with nondecreasing weights it is convex and contains the maximum, the range and the Gini mean difference as special cases. In location theory the same idea became the ordered median problem (Nickel & Puerto, 2005), with linear programming representations of sorted sums through top-k sums (Ogryczak & Śliwiński, 2003; Ogryczak & Tamir, 2003) and a theory of equitable location built on inequality measures (Ogryczak, 2000). Recent work extends the machinery to equity within facilities (Blanco, Marín, & Puerto, 2024), to Benders decomposition for the discrete ordered median problem (Ljubić, Pozo, Puerto, & Torrejón, 2024), to quadratic ordered measures such as the variance (Blanco et al., 2026), and to the ordered median travelling salesman problem, in which the edges of a single tour are weighted by rank (Ljubić, Marín, Puerto, & Temprano, 2026). The broader inequity-averse literature is surveyed by Karsu and Morton (2015), and Tsang and Shehadeh (2026) characterise a class of convex fairness measures and show how stable optimal solutions are with respect to the choice of measure. In all of these works the ordered measure is applied to a deterministic vector. Our Theorem 5 gives the expectation of an ordered measure of independent random variables in closed integral form, and Proposition 7 its derivative with respect to each component; we are not aware of an analogue in the ordered-median literature.

2.3 Fairness under uncertainty

Only a handful of papers combine routing, fairness and uncertainty, and none of them concerns the fairness of drivers' random workloads. Kinable, Sluijk, Gendreau, Rei, and Van Woensel (2026) study a stochastic vehicle routing problem with partial deliveries in which demand is random and fairness is defined among *customers*, as a minimum expected fill rate with a Rawlsian criterion; the quantity to be shared is the service received, not the work done. Gao, Iravani, and Sinclair (2026) combine sequential fair allocation with routing under demand uncertainty

and define max-min objectives over locations, both ex post (per realisation) and in a forward sense; again the recipients of fairness are locations, and the measure is max-min rather than a general ordered measure. On the workforce side, Anh (2022) assign tasks to workers with uncertain service times so that the difference in total working time stays below a threshold with high probability, using a distributionally robust joint chance constraint approximated by the worst-case conditional value-at-risk (CVaR); there is no routing. Lin, Bakker, and Leung (2022) design two-echelon routes for meal delivery with a workload-fairness criterion among workers and protect it against service-time uncertainty with robust constraints; the fairness measure is evaluated on the deterministic side of the robust counterpart, not as a distribution of realised workloads. Gig-economy and rideshare work treats fairness of assignment *probabilities* and of matching under stochastic arrivals (Nanda, Xu, Sankararaman, Dickerson, & Srinivasan, 2020; Singh, Das, & Chakraborty, 2023) and fair allocation of delivery tasks (Basik, Gedik, Ferhatosmanoglu, & Wu, 2021; Hosseini & Schierreich, 2025); the object is an assignment, not the duration of a route.

The word *consistency* is used differently in the consistent vehicle routing problem, where it means that customers are visited by the same driver at similar times across days (Groër, Golden, & Wasil, 2009; Kovacs, Golden, Hartl, & Parragh, 2015). This notion has been extended to stochastic customers and demands with a two-stage model solved by sample average approximation and decomposition (Alvarez, Cordeau, & Jans, 2024), to robust demand (Wu, Jin, & Bo, 2025) and to driver equity with flexible consistency (Yu, Hu, & Wu, 2024). It is unrelated to the route consistency of Ljubić et al. (2026) and of this paper, which requires each route to be an optimal tour of its own customers.

The distinction between ex-ante and ex-post fairness that organises our model is well established outside routing. In welfare economics, Fleurbaey and Peragine (2012) contrast equality of prospects with equality of outcomes. Gutjahr (2021) formalises ex-ante and ex-post inequity aversion in stochastic decision processes and relates them to time consistency, and Mostajabdaveh, Gutjahr, and Sibel Salman (2019) combine ex-ante and ex-post Gini-based inequity in stochastic shelter location, followed by Alem, Caunhye, and Moreno (2022) for two-stage humanitarian location–allocation and by Mostajabdaveh, Salman, and Gutjahr (2025) for a bi-objective relief routing problem with a Gini index of unsatisfied demand. Sinclair, Jain, Banerjee, and Yu (2023) characterise the envy–efficiency trade-off of sequential allocation under stochastic demand, Shehadeh (2023) reduce disparities in transport distance in stochastic facility location, Ye, Hanasusanto, and Xie (2024) define distributional fairness through Wasserstein distances, and Barak and Talgam-Cohen (2026) use exactly our two definitions (the measure of expected loads against the expected measure) in facility-location mechanism design. These works establish that the two notions differ and that the ex-post one is the relevant one for affected individuals; none of them concerns routing or ordered measures of random route durations. We bring the distinction to fair routing, quantify the gap between the two notions (Proposition 9) and show how to optimise the ex-post notion exactly in the objective.

2.4 Stochastic travel times in vehicle routing

Routing with random travel times has a rich literature, but its objectives are cost, feasibility or reliability rather than equity. Laporte, Louveaux, and Mercure (1992) introduce a priori routes with chance-constrained and recourse models, Kenyon and Morton (2003) include a makespan-type objective on the expected or quantile completion time of the longest route, and Bakach, Campbell, Ehmke, and Urban (2021) study makespan objectives with correlated travel times; these are the closest to our maximum measure, but they optimise a moment or quantile of the makespan, not an ordered measure of all routes, and they do not consider the dispersion of workloads. Distribution fitting and correlation are discussed by Gómez, Mariño, Akhavan-Tabatabaei, Medaglia, and Mendoza (2016), stochastic and robust formulations with deadlines by Adulyasak and Jaillet (2016), a priori optimisation with recourse for time windows by Errico, Desaulniers, Gendreau, Rei, and Rousseau (2016, 2018), the cost–service trade-off by J. Zhang, Lam, and Chen (2013), service-level guarantees by Ehmke, Campbell, and Urban (2015), and CVaR-type risk in latency location-routing by Osorio-Mora, Saldanha-da Gama, and Toth (2025); the surveys of Oyola, Arntzen, and Woodruff (2018) and Gendreau, Jabali, and Rei (2016) give the wider picture. Correlated travel times are handled exactly by Rostami, Desaulniers, Errico, and Lodi (2021) and approximately by Bomboi, Buchheim, and Pruente (2021); Rajabi-Bahaabadi, Shariat-Mohaymany, Babaei, and Vigo (2019); robust and distributionally robust variants include Han, Lee, and Park (2014), Jaillet, Qi, and Sim (2016), D. Zhang, Li, Sun, and Hou (2021) and Ghosal and Wiesemann (2020). Sample average approximation (Verweij, Ahmed, Kleywegt, Nemhauser, & Shapiro, 2003) is the generic alternative to our closed-form evaluation, and we compare against it directly (E15). Mean–risk path choice (Nikolova & Stier-Moses, 2014; S. Sen, Pillai, Joshi, & Rathi, 2001) motivates our definition of a risk-consistent tour as one that minimises the driver's mean plus a multiple of the standard deviation. Conditional value-at-risk (R. Rockafellar & Uryasev, 2002; R. T. Rockafellar & Uryasev, 2000), which we report for the realised range, and the price of fairness (Bertsimas, Farias, & Trichakis, 2011; Mo & Walrand, 2000), which E9 traces for stochastic fairness, complete the background.

2.5 Order statistics and parametric optimisation

Our evaluator rests on classical results on order statistics of independent, non-identically distributed variables. Their joint distribution can be written with permanents (Bapat & Beg, 1989; Vaughan & Venables, 1972), which are expensive to evaluate (Balakrishnan, 2007; Glueck, Karimpour-Fard, Mandel, Hunter, & Muller, 2008); the alternative we use expresses the event "at most r variables lie below t" through the Poisson–binomial distribution of the number of successes in independent non-identical trials (Hoeffding, 1956), whose probability mass function is computed exactly by a short recursion (Biscarri, Zhao, & Brunner, 2018; Fernandez & Williams, 2010; Hong, 2013); see also David and Nagaraja (2003) and P. K. Sen (1970). To our knowledge the Poisson–binomial representation has not been used to evaluate or differentiate ordered fairness measures in optimisation. For the lower level, supported efficient solutions of bi-criteria problems are enumerated by dichotomic (weighted-sum) search (Aneja & Nair, 1979), see Ehrgott (2005); Ehrgott and Gandibleux (2000); Ralphs, Saltzman, and Wiecek (2006), and

Table 1: Positioning of SORTIE among the closest works. Drivers: fairness among drivers' workloads; Ordered: general ordered measures; Random durations: route durations are random; Ex post: fairness of realised outcomes is optimised; Consistency: every route is an optimal tour of its customers. (○: partly, see text.)

Work	Routing	Drivers	Ordered	Random durations	Ex post	Consistency
Matl et al. (2018)	✓	✓	○			
van Rossum et al. (2026b)	✓	✓	○			✓
Ljubić et al. (2026)	✓	✓	✓			✓
Bakach et al. (2021); Kenyon and Morton (2003)	✓	○		✓	○	
Kinable et al. (2026)	✓					
Gao et al. (2026)	✓				✓	
Lin et al. (2022)	✓	✓		○		
Anh (2022)		✓		✓	○	
Gutjahr (2021); Mostajabdaveh et al. (2019)			○		✓	
SORTIE (this paper)	✓	✓	✓	✓	✓	✓

small travelling salesman problems are solved by the Held–Karp recursion (Bellman, 1962; Held & Karp, 1962).

3 Model

Let $G = (V, A)$ be a complete graph with depot 0, customers $V' = \{1, \dots, n\}$, symmetric costs c_{ij} satisfying the triangle inequality, demands q_i and m identical vehicles of capacity Q. A plan consists of a partition $S_1, \dots, S_m$ of V' into nonempty capacity-feasible sets and a tour r_k of $\{0\} \cup S_k$ for each vehicle.

Random workloads. The travel time of arc (i, j) is $\tau_{ij} = Y c_{ij} \varepsilon_{ij}$, where $Y = \exp(s_Y Z - s_Y^2/2)$, $Z \sim N(0, 1)$, is a fleet-wide day factor with $\mathbb{E}Y = 1$, and the arc disturbances ε_{ij} are independent of each other and of Y with mean 1 and variance v^2 (gamma in our experiments). The workload of driver k is $\Phi_k = Y X_k$ with

$$X_k = \sum_{(i,j)\in r_k} c_{ij}\varepsilon_{ij}, \qquad \mathbb{E}X_k = L_k = \sum_{(i,j)\in r_k} c_{ij}, \qquad \operatorname{Var} X_k = \sigma_k^2 = v^2 W_k, \quad W_k = \sum_{(i,j)\in r_k} c_{ij}^2.$$

L_k is the tour length and W_k its *quadratic length*. Given Y the workloads are independent; unconditionally they are positively correlated.

Ordered fairness. For $a \in \mathbb{R}^m$ with order statistics $a_{(1)} \le \cdots \le a_{(m)}$ and weights $\lambda \in \mathbb{R}^m$, $\ell_\lambda(a) = \sum_j \lambda_j a_{(j)}$. We use the maximum ($\lambda = e_m$), the range ($e_m - e_1$), the GMD ($\lambda_j = 2(2j - m - 1)/m^2$) and the mean absolute deviation from the median (MAD; $\lambda = (-1, \dots, -1, 0, 1, \dots, 1)/m$ for odd m), together with the empirical variance $\operatorname{Var}_{\text{emp}}(a) = m^{-1} \sum_k (a_k - \bar{a})^2$. All four λ are nondecreasing.

Definition 1 (Three notions of fairness)**.** For a plan with random workloads Φ: *ex-ante* unfairness $\ell_\lambda(\mathbb{E}\Phi)$; *ex-post* unfairness $\mathbb{E}\ell_\lambda(\Phi)$; *risk of unfairness* $\operatorname{CVaR}_\beta \ell_\lambda(\Phi)$.

A deterministic fair model optimises the ex-ante notion. Drivers experience the ex-post one. SORTIE optimises ex-post unfairness and reports the CVaR out of sample.

4 Theory

This section develops the properties of expected ordered fairness that the model and the algorithm rely on. Section 4.1 collects elementary properties of ordered measures; Section 4.2 shows that fleet-wide shocks drop out of the objective; Section 4.3 gives the exact evaluation formula and Section 4.4 its derivatives; Section 4.5 quantifies the fairness illusion and shows that it is a consequence of balance itself; Section 4.6 treats route consistency; and Section 4.7 states the SORTIE problem and its complexity. Throughout, $X = (X_1, \dots, X_m)$ is a vector of route durations, $L = \mathbb{E}X$, $\sigma_k^2 = \operatorname{Var} X_k$, and $\lambda \in \mathbb{R}^m$.

4.1 Ordered measures

Lemma 2 (Properties of ordered measures)**.** *For $a, b \in \mathbb{R}^m$, $y > 0$ and $c \in \mathbb{R}$:*

(a) $\ell_\lambda(ya) = y\,\ell_\lambda(a)$ *and* $\ell_\lambda(a + c\mathbf{1}) = \ell_\lambda(a) + c\sum_j \lambda_j$*;*

(b) $|\ell_\lambda(a) - \ell_\lambda(b)| \le \|\lambda\|_1 \|a - b\|_\infty$*;*

(c) if λ is nondecreasing, $\ell_\lambda(a) = \max_\pi \sum_k \lambda_{\pi(k)} a_k$ over all permutations π, so ℓ_λ is convex, and the maximum is attained by the permutation that assigns to each component its rank in a.

Proof. (a) Sorting commutes with multiplication by $y > 0$ and with adding a constant to every component. (b) The sorted vectors satisfy $|a_{(j)} - b_{(j)}| \le \|a - b\|_\infty$ for every j. (c) This is the rearrangement inequality: for nondecreasing λ the sum $\sum_k \lambda_{\pi(k)} a_k$ is largest when larger weights meet larger components. A maximum of linear functions is convex. □

The maximum, range, GMD and MAD all have nondecreasing weights; the range, GMD and MAD in addition have $\sum_j \lambda_j = 0$ (*dispersion measures*), so by (a) they are invariant to a common additive shift, while the maximum has $\sum_j \lambda_j = 1$.

4.2 Common shocks

Proposition 3 (Common shocks)**.** *For every λ,*

$$\mathbb{E}\ell_\lambda(\Phi) = \mathbb{E}\ell_\lambda(X) \qquad \textit{and} \qquad \mathbb{E}\operatorname{Var}_{\mathrm{emp}}(\Phi) = e^{s_Y^2}\,\mathbb{E}\operatorname{Var}_{\mathrm{emp}}(X).$$

Proof. By Lemma 2(a), $\ell_\lambda(yX) = y\ell_\lambda(X)$; take expectations using the independence of Y and X and $\mathbb{E}Y = 1$. The empirical variance is homogeneous of degree two and $\mathbb{E}Y^2 = e^{s_Y^2}$. □

Corollary 4 (Additive shocks)**.** *If $\Phi = X + Z\mathbf{1}$ for a random variable Z with finite mean, and $\mathbb{E}\ell_\lambda(X)$ exists (for example, a fleet-wide delay at the depot), then $\mathbb{E}\ell_\lambda(\Phi) = \mathbb{E}\ell_\lambda(X) + \mathbb{E}Z\sum_j \lambda_j$. In particular every dispersion measure is unaffected, and the expected maximum shifts by $\mathbb{E}Z$ for every plan alike.*

Proof. Lemma 2(a) and linearity of expectation. □

Fleet-wide shocks, multiplicative or additive, therefore do not change the ranking of plans by expected ordered fairness, which reduces evaluation to the conditionally independent vector X. They do change the distribution of unfairness: Proposition 3 does not extend to quantiles or to the CVaR, which grows by 58% at $s_Y = 0.5$ in our experiments (E11).

4.3 Exact evaluation

Theorem 5 (Poisson–binomial order-statistic integral)**.** *Let $X_1, \dots, X_m \geq 0$ be independent with CDFs F_k. For $t \geq 0$ let $N(t) = \#\{k : X_k \leq t\}$, a Poisson–binomial count with success probabilities $F_k(t)$, and $\pi_r(t) = \mathbb{P}(N(t) = r)$. With tail sums $\Lambda_r = \sum_{j=r+1}^{m} \lambda_j$,*

$$\mathbb{E}\ell_\lambda(X) = \int_0^\infty \sum_{r=0}^{m-1} \Lambda_r \, \pi_r(t) \, dt. \tag{1}$$

Proof. $X_{(j)} > t$ if and only if fewer than j of the X_k are at most t, so $\mathbb{E}X_{(j)} = \int_0^\infty \mathbb{P}(N(t) \leq j-1)\, dt$. Hence $\mathbb{E}\ell_\lambda(X) = \int_0^\infty \sum_j \lambda_j \sum_{r \leq j-1} \pi_r(t)\, dt = \int_0^\infty \sum_r \pi_r(t) \sum_{j \geq r+1} \lambda_j \, dt$, where Tonelli's theorem is applied to the positive and negative parts of λ separately. □

Corollary 6 (Maximum, minimum and range)**.** *Under the assumptions of Theorem 5, and if the means are finite,*

$$\mathbb{E}\max_k X_k = \int_0^\infty \Big(1 - \prod_k F_k(t)\Big) dt, \qquad \mathbb{E}\min_k X_k = \int_0^\infty \prod_k \big(1 - F_k(t)\big) dt,$$

$$\mathbb{E}\big(\max_k X_k - \min_k X_k\big) = \int_0^\infty \Big(1 - \prod_k F_k(t) - \prod_k \big(1 - F_k(t)\big)\Big) dt.$$

Proof. For the maximum, $\lambda = e_m$, so $\Lambda_r = 1$ for $r < m$ and $\sum_{r<m} \pi_r(t) = 1 - \pi_m(t) = 1 - \prod_k F_k(t)$. For the minimum, $\lambda = e_1$, so only $\Lambda_0 = 1$ is non-zero and $\pi_0(t) = \prod_k (1 - F_k(t))$. The range is the difference. □

For the maximum and the range only products of marginals are needed; for general weights, and in particular for the GMD and MAD, all $m+1$ Poisson–binomial probabilities are required. The vector $\pi(t)$ is the coefficient vector of the generating function $\prod_k (1 - F_k(t) + F_k(t)z)$ and is obtained by m convolution steps in $O(m^2)$. Below the smallest lower quantile of the marginals the integrand equals Λ_0, above the largest upper quantile it vanishes ($\Lambda_m := 0$), and in between we use Gauss–Legendre panels whose breakpoints are quantiles of *every* marginal, so that a short, low-variance route is resolved even when the other routes are long (Section 5.2). We use moment-matched gamma marginals $X_k \sim \Gamma(L_k^2/\sigma_k^2, \sigma_k^2/L_k)$. The identity behind (1) is classical for order statistics of non-identical variables (Balakrishnan, 2007; Bapat & Beg, 1989); our contribution is to make it an exact planning objective for ordered fairness, where the tail sums Λ_r absorb the negative weights of dispersion measures, and to differentiate it.

4.4 Rank prices

Proposition 7 (Rank prices)**.** *Let the X_k have densities, let R_k be the rank of X_k and define $\varpi_k = \sum_j \lambda_j \mathbb{P}(R_k = j)$, where $\mathbb{P}(R_k = j) = \int f_k(t)\, \mathbb{P}(N_{-k}(t) = j - 1)\, dt$ and N_{-k} counts the*

other drivers. Then $\frac{d}{d\delta}\mathbb{E}\ell_\lambda(X+\delta e_k)\big|_{\delta=0} = \varpi_k$ *and* $\sum_k \varpi_k = \sum_j \lambda_j$.

Proof. ℓ_λ is piecewise linear with $\partial\ell_\lambda/\partial a_k = \lambda_{R_k(a)}$ off the null set of ties and is $\|\lambda\|_1$-Lipschitz, so differentiation and expectation commute by dominated convergence. The event $R_k = j$ occurs when $X_k = t$ and exactly $j-1$ of the other variables are below t, which gives the integral. The ranks form a permutation, so $\sum_k \mathbb{P}(R_k = j) = 1$. □

Proposition 8 (Convexity and first-order bound)**.** *Under the assumptions of Proposition 7 (independent durations with densities and finite means), let* λ *be nondecreasing and* $F(\delta) = \mathbb{E}\ell_\lambda(X+\delta)$ *for* $\delta \in \mathbb{R}^m$*. Then* F *is convex and differentiable with* $\nabla F(0) = \varpi$*, so that for every shift* δ

$$F(\delta) \;\geq\; F(0) + \sum_k \varpi_k\,\delta_k, \qquad \text{and} \qquad \lambda_1 \leq \varpi_k \leq \lambda_m \quad \forall k.$$

If the means L_k *are pairwise distinct and* $\sigma_k \to 0$ *for all* k*, then* $\varpi_k \to \lambda_{r_k}$*, where* r_k *is the rank of* L_k *in* L*.*

Proof. $\delta \mapsto \ell_\lambda(X+\delta)$ is convex for every realisation (Lemma 2(c)), so F is convex; differentiability and the gradient follow from Proposition 7 applied at every δ. A differentiable convex function lies above its tangent plane. ϖ_k is a convex combination of the λ_j, which lie between λ_1 and λ_m. As $\sigma \to 0$, $X \to L$ in probability and $\mathbb{P}(R_k = r_k) \to 1$ because the means are distinct. □

Proposition 8 explains the role of the rank prices in the search. For a pure shift of route durations, the linear prediction $\sum_k \varpi_k\delta_k$ is a *lower bound* on the true change, and the error is $o(\|\delta\|)$ (of second order for fixed $\sigma > 0$ and smooth densities; the curvature grows like $1/\sigma$ as the noise vanishes). A move changes the variances as well; the dispersion prices $\psi_k = \partial F/\partial\sigma_k$ account for that to first order. In the deterministic limit the prices reduce to the weight of each route's rank, which is how the deterministic screen of the baseline is built. For the range, the GMD and the MAD the prices sum to zero: burdened drivers carry positive prices and light drivers negative ones, and a negative price is exactly the incentive to lengthen a route. Deterministically the price of the lightest route under the range is -1, and a detour of length δ reduces the range by δ; stochastically the incentive is spread over all drivers who might be the lightest on a given day.

4.5 The fairness illusion

Proposition 9 (Fairness illusion)**.** *If* λ *is nondecreasing then*

$$0 \leq \mathbb{E}\ell_\lambda(X) - \ell_\lambda(L) \leq \|\lambda\|_1\,\mathbb{E}\|X-L\|_\infty \leq \|\lambda\|_1\, v\Big(\sum_k W_k\Big)^{1/2}.$$

Moreover $\mathbb{E}\,\mathrm{Var}_{\mathrm{emp}}(X) = \mathrm{Var}_{\mathrm{emp}}(L) + \frac{m-1}{m^2}\sum_k \sigma_k^2$*, and for Gaussian marginals* $\mathbb{E}\,\mathrm{GMD}(X) = m^{-2}\sum_{i\neq j}\left[s_{ij}\sqrt{2/\pi}\,e^{-d_{ij}^2/(2s_{ij}^2)} + d_{ij}(1-2\Phi(-d_{ij}/s_{ij}))\right]$ *with* $d_{ij} = L_i - L_j$ *and* $s_{ij}^2 = \sigma_i^2 + \sigma_j^2$*.*

Proof. Convexity (Lemma 2(c)) and Jensen's inequality give the lower bound, Lemma 2(b) the first upper bound, and $\mathbb{E}\max_k|X_k - L_k| \leq (\mathbb{E}\sum_k (X_k-L_k)^2)^{1/2} = v(\sum_k W_k)^{1/2}$ the second. The

variance identity follows from $\mathbb{E}X_k^2 = L_k^2 + \sigma_k^2$ and $\mathbb{E}\bar{X}^2 = \bar{L}^2 + m^{-2}\sum_k \sigma_k^2$; the GMD formula is the mean of a folded normal. □

A deterministic fair plan therefore *overstates* the fairness drivers will experience. The bound grows with the arc noise and with the quadratic lengths: long arcs, not long routes, are what make a balanced plan unbalanced in practice. The bound of Proposition 9 does not depend on how balanced the plan is. The next result shows that the gap itself depends on it sharply: it is exponentially small when the expected lengths are well separated and of the order of the standard deviation when they are equal.

Proposition 10 (Balance amplifies the illusion)**.** *Let $m \geq 2$, $X_k = L_k + \sigma_k Z_k$ with $Z_1, \ldots, Z_m$ independent standard normal, $\bar{\sigma} = \max_k \sigma_k > 0$, and λ nondecreasing.*

(a) If $L_1 = \cdots = L_m$ and $\sigma_1 = \cdots = \sigma_m = \sigma$, then $\mathbb{E}\ell_\lambda(X) - \ell_\lambda(L) = \sigma \sum_j \lambda_j \mu_{j:m}$, where $\mu_{j:m}$ is the expected j-th order statistic of m standard normals; the constant $\sum_j \lambda_j \mu_{j:m}$ is positive unless λ is constant.

(b) If the L_k are pairwise separated by at least $\Delta > 0$, then

$$0 \leq \mathbb{E}\ell_\lambda(X) - \ell_\lambda(L) \leq 2\|\lambda\|_1 \,\bar{\sigma}\sqrt{\tfrac{m(m-1)}{2}}\; e^{-\Delta^2/(8\bar{\sigma}^2)}.$$

Proof. (a) With equal means, $\ell_\lambda(X) = \ell_\lambda(L) + \sigma\ell_\lambda(Z)$ by Lemma 2(a), and $\mathbb{E}\ell_\lambda(Z) = \sum_j \lambda_j \mu_{j:m}$. The $\mu_{j:m}$ are strictly increasing in j and sum to zero by symmetry, so by Chebyshev's sum inequality $\sum_j \lambda_j \mu_{j:m} \geq \frac{1}{m}\sum_j \lambda_j \sum_j \mu_{j:m} = 0$, with equality only if λ is constant. (b) Let r be the ranking of L and A the event that X is not ranked like L. By Lemma 2(c), $\ell_\lambda(X) = \max_\pi \sum_k \lambda_{\pi(k)} X_k$ and $\ell_\lambda(L) = \sum_k \lambda_{r(k)} L_k$, so

$$G := \ell_\lambda(X) - \ell_\lambda(L) - \sum_k \lambda_{r(k)}\sigma_k Z_k = \max_\pi \sum_k (\lambda_{\pi(k)} - \lambda_{r(k)})(L_k + \sigma_k Z_k).$$

On A^c the maximum is attained at $\pi = r$ and $G = 0$. Since r maximises $\sum_k \lambda_{\pi(k)} L_k$, the L-part of every term is non-positive, so $G \leq \bar{\sigma}\max_k |Z_k| \max_\pi \sum_k |\lambda_{\pi(k)} - \lambda_{r(k)}| \leq 2\|\lambda\|_1 \bar{\sigma}\max_k |Z_k|$. The correction term has mean zero, hence $\mathbb{E}\ell_\lambda(X) - \ell_\lambda(L) = \mathbb{E}[G\,\mathbf{1}_A] \leq 2\|\lambda\|_1 \bar{\sigma}\,(\mathbb{E}\max_k Z_k^2)^{1/2}\mathbb{P}(A)^{1/2}$ by Cauchy–Schwarz, with $\mathbb{E}\max_k Z_k^2 \leq m$. If the ranking of X differs from that of L, some pair of routes adjacent in L is reversed; for such a pair (k, l) with $L_l - L_k \geq \Delta$, reversal requires $\sigma_k Z_k - \sigma_l Z_l \geq \Delta$, which has probability at most $\bar{\Phi}(\Delta/(\sqrt{2}\bar{\sigma})) \leq \frac{1}{2}e^{-\Delta^2/(4\bar{\sigma}^2)}$. A union bound over the $m - 1$ adjacent pairs completes the proof. □

Proposition 10 identifies the mechanism behind the size of the illusion: a deterministic fairness objective drives the expected lengths together, and the closer they are, the more the realised ranking is decided by noise rather than by the plan. In the limit of a perfectly balanced plan the gap is a fixed multiple of the standard deviation (for the range with $m = 5$ the constant is $\mu_{5:5} - \mu_{1:5} \approx 2.33$), whereas for a plan whose routes differ by several standard deviations it is negligible. Fair plans are therefore exactly the plans in which the deterministic assessment is least reliable. The data agree: on the 25 main-grid instances the deterministic value of the *cost-optimal* plans, whose route lengths differ by about 110 distance units between neighbours

(median), understates the true range by 5.5% on average, against 23% for the Det range plans at $B = 1.10$, and the same pattern holds for the maximum (4.6% against 9%), the GMD (4.8% against 19%) and the MAD (4.3% against 20%).

4.6 Route consistency under uncertainty

Proposition 11 (Monotonicity no longer protects drivers)**.** *With deterministic workloads, every monotone measure ($\lambda \geq 0$) has an optimal plan made of TSP-optimal routes (Matl et al., 2018). With random workloads this fails even for $\mathbb{E}\max$.*

Example 12. Two drivers; $X_1 \equiv 100$. Driver 2 can drive tour A ($L = 100$, $\sigma = 10$) or tour B ($L = 101$, $\sigma \approx 0$). With Gaussian X_2, $\mathbb{E}\max = 100 + 10\varphi(0) \approx 103.99$ under A and ≈ 101 under B, so the optimiser picks the longer tour B.

The example generalises: by Corollary 6, $\mathbb{E}\max$ depends on the whole distribution of each route, and lowering a route's variance can reduce the expected maximum by more than the extra length it costs. Route consistency must therefore be imposed for every measure, and it needs a risk attitude. We let each driver rank the tours of their customer set by

$$\rho_\kappa(r) = \frac{\mathbb{E}[\Phi(r) \mid Y] + \kappa \operatorname{SD}[\Phi(r) \mid Y]}{Y} = L(r) + \kappa v \sqrt{W(r)},$$

which ranks tours identically on every day. $\kappa = 0$ is TSP-optimality (ties broken by smaller W).

Theorem 13 (Risk-consistent tours are supported)**.** *For $\kappa \geq 0$, some minimiser of ρ_κ over the tours of a fixed customer set is a supported extreme tour: it minimises $L(r) + \theta W(r)$ for some $\theta \in [0, \infty)$, or minimises W lexicographically after L ($\theta = \infty$). All supported extreme tours are enumerated by dichotomic search, each step being one Held–Karp run with arc weights $c_{ij} + \theta c_{ij}^2$.*

Proof. $f(L, W) = L + \kappa v \sqrt{W}$ is concave and nondecreasing on $\mathbb{R}^2_+$. A concave function bounded below on a pointed polyhedron attains its minimum at an extreme point; since f is nondecreasing, the minimum of f over the finite set $P = \{(L(r), W(r))\}$ equals its minimum over $\operatorname{conv}(P) + \mathbb{R}^2_+$ and is attained at an extreme point of that set. These extreme points are exactly the unique minimisers of $w_1 L + w_2 W$ with $(w_1, w_2) \geq 0$, $(w_1, w_2) \neq 0$ (for $w_1 = 0$ with ties broken by smaller L), and dichotomic search enumerates all of them (Aneja & Nair, 1979). Tours in the relative interior of a hull edge are dominated by the edge's end points and need not be enumerated. With heterogeneous noise (ε_{ij} with variance $v^2 H_{ij}^2$) the same holds with $W = \sum (H_{ij} c_{ij})^2$ and arc weights $c_{ij} + \theta H_{ij}^2 c_{ij}^2$. □

Corollary 14 (Effort and monotonicity in κ)**.** *(a) If a customer set has $s \geq 2$ supported extreme tours with unique minimisers of every weighted problem, the dichotomic search solves $2s - 1$ weighted tour problems (two if $s = 1$), counting the two lexicographic end-point problems. (b) Let $v > 0$, and let r_κ and $r_{\kappa'}$ minimise ρ_κ and $\rho_{\kappa'}$ with $\kappa < \kappa'$. Then $W(r_{\kappa'}) \leq W(r_\kappa)$ and $L(r_{\kappa'}) \geq L(r_\kappa)$: more risk-averse drivers choose tours that are longer but less variable.*

Proof. (a) Two problems give the end points; every later problem either produces a new supported tour (at most $s - 2$ times) or certifies one of the $s - 1$ edges between adjacent supported

tours. (b) Write $u = v\sqrt{W}$. Optimality gives $L_\kappa + \kappa u_\kappa \le L_{\kappa'} + \kappa u_{\kappa'}$ and $L_{\kappa'} + \kappa' u_{\kappa'} \le L_\kappa + \kappa' u_\kappa$; adding yields $(\kappa' - \kappa)(u_{\kappa'} - u_\kappa) \le 0$, so $u_{\kappa'} \le u_\kappa$, and then the first inequality gives $L_{\kappa'} \ge L_\kappa$. □

Under homogeneous noise the length and the quadratic length of the tours of a fixed customer set are nearly comonotone, so s is small and the risk-consistent tour rarely differs from the shortest one; E13 confirms this (at most 16% of the routes change even at $\kappa = 4$).

4.7 The SORTIE problem

$$
\begin{aligned}
\min\ & \mathbb{E}\,\ell_\lambda(X(r_1, \dots, r_m)) \\
\text{s.t.}\ & (S_1, \dots, S_m) \text{ capacity-feasible partition},\ S_k \neq \emptyset, \qquad \textstyle\sum_k L(r_k) \le B \cdot C^*, \\
& r_k \in \arg\min\{\rho_\kappa(r) : r \text{ a tour of } \{0\} \cup S_k\} \quad \forall k,
\end{aligned}
$$

where C^* is the minimum expected total length and $B \ge 1$ the budget factor. With $v = 0$ and $\kappa = 0$ this is the X-CVRP-TSP of Ljubić et al. (2026).

Proposition 15 (Complexity)**.** *For every noise level $v \ge 0$, the SORTIE problem with the maximum measure and $\kappa = 0$ is NP-hard, and so is the lower level for a single customer set.*

Proof. Take $m = 1$, any $B \ge 1$ and no capacity limit (with one vehicle the budget holds automatically). Then the only partition is $S_1 = V'$, $\mathbb{E}\max_k X_k = \mathbb{E}X_1 = L(r_1)$, and r_1 must be a shortest tour of $\{0\} \cup V'$: solving the problem, or its lower level, solves the travelling salesman problem. □

The problem is therefore at least as hard as the deterministic fair routing problem of Ljubić et al. (2026), which it contains, and the stochastic objective adds a non-linear evaluation on top. This motivates the heuristic of Section 5, which keeps the lower level exact on accepted plans (Proposition 16) and evaluates the objective exactly by Theorem 5.

5 The SORTIE algorithm

The SORTIE problem combines three difficulties: a capacitated partitioning problem in the upper level, a travelling salesman problem (or, for $\kappa > 0$, a bi-criteria tour problem) for every route in the lower level, and an objective that is an expectation over order statistics of random variables. Exact methods for the deterministic special case (Ljubić et al., 2026; van Rossum et al., 2026b) report instances with up to about 25 customers (van Rossum et al., 2026b); the stochastic objective is non-linear in the route statistics and is not representable in their formulations. We therefore propose a bilevel local search that keeps the lower level exact wherever it is applied to a plan the search accepts, and uses the theory of Section 4 in two places: PBOI evaluates the objective without sampling, and the rank prices predict the effect of a move before it is evaluated. This section describes the representation and lower level (§5.1), the evaluator (§5.2), the first-order screen (§5.3), the exact confirmation that keeps every accepted plan bilevel feasible (§5.4), the ruin-and-recreate perturbation and acceptance rule (§5.5), and the parameters and complexity (§5.6). Figure 1 gives an overview and Algorithm 1 the pseudocode; a detailed pseudocode of the improvement step and the full parameter table are in the Online Supplement.

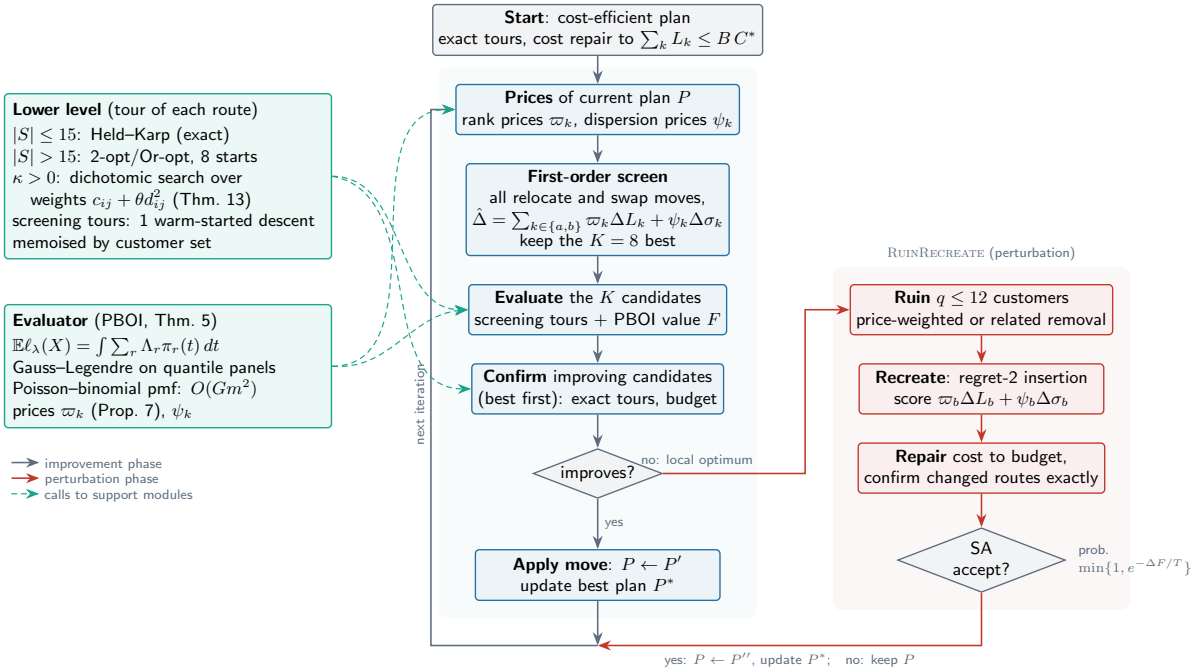


Figure 1: The SORTIE algorithm. The improvement phase (blue) screens every relocate and swap move with the rank and dispersion prices, evaluates the K best with screening tours and PBOI, and confirms improving candidates with the exact lower level. When no confirmed move improves the plan, the perturbation phase (red) removes and reinserts customers guided by the same prices, repairs the budget, confirms the changed routes and accepts by simulated annealing. The two support modules (green) supply tours and objective values; dashed arrows mark their calls.

5.1 Representation and lower level

A solution is a list of m customer sequences, one per vehicle. The upper level is the partition; the sequence of each route is not a decision of the upper level but the output of the lower-level solver applied to the route's customer set S. For $\kappa = 0$ the lower level returns a shortest tour of $\{0\} \cup S$, with ties broken by smaller quadratic length. For $|S| \le 15$ it is solved exactly by the Held–Karp recursion (Held & Karp, 1962) compiled with numba, in $O(2^{|S|}|S|^2)$ time (about 5 ms at 15 customers); for larger sets we use 2-opt and Or-opt (segments of one to three customers, both orientations) from eight starts: a warm start built from the route's previous sequence by cheapest insertion of new customers, the nearest-neighbour tour, and random permutations. For $\kappa > 0$ the lower level minimises $\rho_\kappa = L + \kappa v\sqrt{W}$ by the dichotomic search of Theorem 13: starting from the tour minimising L (ties by W) and the tour minimising W (ties by L), it repeatedly solves the weighted problem with arc weights $c_{ij} + \theta d_{ij}^2$, where θ is the slope between two known supported tours and $d_{ij} = H_{ij}c_{ij}$ (so $d = c$ under homogeneous noise), until no new supported tour appears; every weighted problem is solved like the $\kappa = 0$ case. Lower-level results are memoised by customer set, so a set that reappears during the search is never solved twice.

Algorithm 1 SORTIE: main loop

Require: instance, measure λ, noise level v, risk weight κ, budget $B\,C^*$, time limit

$P \leftarrow$ cost-efficient plan; re-solve every route with the exact lower level; repair cost if $\sum_k L_k > B\,C^*$
$P^* \leftarrow P$; $T_0 \leftarrow 0.02\,|\mathbb{E}\ell_\lambda(X(P))|$; *stuck*$\leftarrow$NONE
while time remains **do**
  compute rank prices ϖ and dispersion prices ψ of P (Prop. 7)
  if *stuck*$\neq P$ **then**
    $P' \leftarrow$ImproveStep(P, ϖ, ψ) ▷ screen, evaluate, confirm (§5.3–5.4)
    if $P' \neq$NONE **then** $P \leftarrow P'$; update P^*; **continue**
    else *stuck*$\leftarrow P$
    end if
  end if
  $P'' \leftarrow$RuinRecreate(P, ϖ, ψ); repair cost; confirm changed routes exactly
  accept P'' with probability $\min\{1, \exp(-(F(P'') - F(P))/T)\}$, $T = T_0\,(10^{-3})^{\text{elapsed/limit}}$
  update P^*
end while
return P^*

5.2 Evaluation

Given the tour lengths L_k and quadratic lengths W_k, every route has mean L_k and standard deviation $\sigma_k = v\sqrt{W_k}$, and its duration is modelled by a gamma marginal with these two moments (normal marginals are an option; E16 shows that the choice hardly matters). The objective $F(P) = \mathbb{E}\ell_\lambda(X)$ is evaluated by the integral of Theorem 5. The integrand is smooth between the quantiles of the marginals and changes quickly near them, so we integrate by six-point Gauss–Legendre rules on the panels delimited by the quantiles of every marginal at 13 levels from 10^{-12} to $1-10^{-12}$; with $m = 5$ this gives about 360 nodes. At each node the Poisson–binomial probability mass function of the number of routes finished by time t is computed by the standard recursion over routes in $O(m^2)$, so one evaluation costs $O(Gm^2)$ for G nodes, about 0.4 ms. The rank prices of Proposition 7 reuse the same nodes, with the mass function of the other $m-1$ routes, in $O(Gm^3)$; the dispersion prices $\psi_k = \partial F/\partial\sigma_k$ are obtained by forward differences with step $0.05\,\sigma_k$, i.e. m further evaluations. For the expected empirical variance all quantities are in closed form (Proposition 9). Prices are computed once per accepted plan.

5.3 First-order screening

The neighbourhood consists of all *relocate* moves (customer i from route a to route b) and all *swap* moves (customers $i \in a$ and $j \in b$ exchanged), about $n(m-1) + n^2(m-1)/(2m)$ moves in total. Evaluating every move exactly would require two lower-level solves and one PBOI evaluation per move. Instead, for every route we precompute, in vectorised form, the change in L and W caused by removing each of its customers (bridging its neighbours), by inserting any customer at its cheapest position, and by replacing each of its customers by any other customer at the cheapest position in the shortened route. These insertion deltas approximate the change

of the route statistics, and the change of the objective is predicted to first order by the prices:

$$\hat{\Delta} = \sum_{k \in \{a,b\}} \Big(\varpi_k \, \Delta L_k + \psi_k \, v\big(\sqrt{W_k + \Delta W_k} - \sqrt{W_k}\big) \Big). \tag{2}$$

Moves that violate capacity, or whose predicted total length exceeds the budget by more than $\epsilon_B = 2\%$, are discarded; the remaining moves are ranked by $\hat{\Delta}$ and the best $K = 8$ are evaluated. By Proposition 7, ϖ_k is the exact derivative of F with respect to a shift of route k, so (2) is the linearisation of F in the route statistics, with the insertion deltas standing in for the re-optimised tours. A negative ϖ_k (possible for the range, the GMD and the MAD) predicts a gain from *lengthening* route k, which is exactly the incentive the lower level neutralises; the screen therefore proposes such moves, and the exact evaluation then reveals whether they survive the re-optimisation of the tour. E14 shows that the prediction ranks moves with a Spearman correlation of about 0.9 with their exact effect. For the deterministic baseline the same screen is used with the deterministic rank prices (λ at the rank of each route, $\psi = 0$), for SAA with the empirical rank frequencies over the scenarios, and for Det-σ with the deterministic prices of $L + \sigma$, so all methods share the search and differ only in the objective.

5.4 Exact confirmation

The K candidates are evaluated with *screening tours*: a single 2-opt/Or-opt descent from the warm start (for $\kappa > 0$ inside the same dichotomic search, capped at 24 tours), without Held–Karp. Improving candidates are then confirmed in order of their screened value: both changed routes are re-solved by the exact lower level, the objective is re-evaluated, and the move is applied only if it still improves and respects the budget. Plans produced by the perturbation are confirmed route by route in the same way.

Proposition 16 (Bilevel feasibility of accepted plans)**.** *Every plan that Algorithm 1 accepts as its current or best plan is capacity feasible, satisfies $\sum_k L_k \le B\,C^*$, and routes every customer set on the tour returned by the exact lower level.*

Proof. The start plan is re-solved by the exact lower level and repaired by cost descent, whose steps are themselves confirmed moves. A plan becomes current only through ImproveStep, which returns a plan only after re-solving the two changed routes exactly and re-checking the budget, or through the perturbation, whose output is accepted only after every route that differs from the current plan has been re-solved exactly and the budget re-checked. All other routes are unchanged and were exact before. Capacity is checked for every move and every insertion. □

The screening tour is never shorter than the exact one, so a move that improves only with exact tours can be missed; this affects the quality of the search, not feasibility. E14 shows that 28 of 30 final SORTIE plans admit no improving relocate or swap move at all, i.e. the search ends in genuine local optima of the neighbourhood. Together with a cap on the ruin size, lazy confirmation raises the iteration rate of SORTIE-0 at $n = 100$ from about 250 to about 1700 per minute, and at $n = 75$ from about 170 to about 1800.

5.5 Perturbation and acceptance

When no screened move improves the current plan, SORTIE applies a ruin-and-recreate perturbation. The number of removed customers q is drawn uniformly from $\{2, \dots, \max(3, \min(12, \lceil n/4 \rceil))\}$. With probability 1/2 the removal pool is *price-weighted*: routes are ordered by sampling without replacement with probabilities proportional to $|\varpi_k|$, and their customers are added to the pool until it holds at least q; otherwise the pool is all customers. From the pool, with probability 1/2 the q customers closest to a random seed customer are removed (related removal), otherwise q random customers; no route is emptied. The removed customers are reinserted by *price-scored regret-2 insertion*: inserting customer i at its cheapest position in route b is scored by

$$s_b(i) = \varpi_b \, \Delta L_b + \psi_b \, \Delta\sigma_b + 10^{-3} \, \Delta L_b,$$

the customer with the largest difference between its best and second-best capacity-feasible score is inserted first, and the scores of the modified route are updated. The small cost term breaks ties in favour of shorter insertions. If the result exceeds the budget it is repaired by cost descent: ImproveStep with unit prices and zero dispersion prices, so that $\hat{\Delta}$ is the change in total length, for at most 200 steps; if the budget cannot be met the perturbation is discarded. The repaired plan is confirmed exactly (§5.4) and accepted by simulated annealing with temperature $T = T_0(10^{-3})^{\tau}$, where τ is the elapsed fraction of the time limit. The initial temperature $T_0 = 0.02\,|\mathbb{E}\ell_\lambda(X(P_0))|$ is computed from the PBOI value of the start plan for every method, so that Det, SAA and SORTIE accept uphill moves of the same absolute size.

5.6 Parameters and complexity

The parameters (listed in the Online Supplement) are the same for all methods and instances; none was tuned per instance. E14 shows that the number of evaluated candidates K between 1 and 32 does not change the median result. One improvement step costs the precomputation of the deltas, $O(m\, n_k^2 N)$ for routes of n_k customers and $N = n + 1$ nodes, which is vectorised; the ranking of all moves, $O(n^2)$; K screening re-solves of two routes, each a local-search descent; K PBOI evaluations, $O(KGm^2)$; and at most a few exact re-solves. The prices cost $O(Gm^3) + m$ PBOI evaluations per accepted plan. On the main grid an iteration takes 7 to 9 ms (median), and the search runs 2300 to 3100 iterations in 15 to 25 s; at $n = 100$ it runs about 3500 iterations in 120 s (E7).

6 Computational study

This section reports the main experiments; the Online Supplement contains further analyses (rank-price diagnostics, noise level, fleet size, day-factor dispersion, drivers' risk attitude, congestion and per-instance results), to which we refer by their labels E1–E17. All code, instances, solved plans and raw results are released (see the data availability statement), and every table and figure is generated from the raw results by the released scripts.

6.1 Set-up

Instances. We follow the slicing rule of the deterministic fair routing literature (Ljubić et al., 2026). Block b of size n consists of the depot and the n consecutive customers $bn+1, \ldots, (b+1)n$ of the CVRPLIB instance X-n641-k35 (Uchoa et al., 2017), with ceiled Euclidean distances, $m = 5$ vehicles and capacity $Q = \lfloor \sum_i q_i/4 \rfloor$, so that one vehicle's worth of slack is available. The main grid uses $n \in \{15, 20, 25, 30, 35\}$ and blocks $b = 0, \ldots, 4$ (25 instances); the scale test uses $n \in \{50, 75, 100\}$ with six blocks each. A second family takes the depot and the first 50 customers of twelve further X instances (X-n101-k25 to X-n153-k22), which keeps their original depot positions and customer distributions (Section 6.6).

Uncertainty and budget. Unless stated otherwise the arc noise has coefficient of variation $v = 0.3$ (gamma) and the day factor $s_Y = 0.15$. C^* is the cost of the best of three restarts of the cost-minimising search, and the total length of every fair plan must satisfy $\sum_k L_k \le B\, C^*$ with $B = 1.10$ unless stated otherwise.

Methods. All methods use the search of Section 5 with the same start plan, random seed and time limit (15 s for $n \le 25$, 25 s for $n \in \{30, 35\}$, 120 s for $n \ge 50$); they differ only in the objective and its prices. *Det* minimises the deterministic measure $\ell_\lambda(L)$, i.e. the model of Ljubić et al. (2026) solved heuristically. *SAA* minimises the sample average of $\ell_\lambda(X)$ over 200 fixed scenarios. *Det-σ* minimises the deterministic measure of the risk-adjusted lengths $L_k + \sigma_k$. *SORTIE-0* minimises the exact ex-post value with shortest tours ($\kappa = 0$) and *SORTIE-1* with risk-consistent tours ($\kappa = 1$).

Evaluation. Every plan is evaluated out of sample by arc-level Monte Carlo simulation with 20 000 scenarios (10 000 for $n \ge 50$), including the day factor, with common random numbers across methods. The reported *ex-post value* is the simulated $\mathbb{E}\ell_\lambda(\Phi)$; improvements are relative to Det, and paired comparisons use the Wilcoxon signed-rank test.

6.2 Accuracy of the evaluator

Table 2 compares PBOI with 50 000-scenario arc-level simulation on 168 plans (cost-efficient and random feasible plans, $n \in \{15, 25, 35, 50\}$). The mean absolute relative error is at most 0.11% for every measure and the worst case 0.44%; the small systematic bias (mean z-scores up to 0.34) comes from the moment-matched gamma marginals. PBOI takes 0.44 ms per plan for $m = 5$, against 0.28 s for 20 000-scenario simulation; a 200-scenario sample average is faster (0.06 ms) but noisy. On ten instances with $n = 10$ and $m = 3$ we also enumerated every capacity- and budget-feasible partition with exact tours: SORTIE-0 found the plan with the smallest exact ex-post value on nine of ten instances (mean gap 0.39%, maximum 3.9%).

6.3 The fairness illusion and the value of ex-post optimisation

Table 3 quantifies Propositions 9 and 10. The value a deterministic fair plan reports understates the true ex-post value by 8% for the maximum, 15–18% for the range, GMD and MAD, and 27% for the variance. The in-sample SAA value is optimistic by about 1% (the optimiser's curse),

Table 2: Accuracy of PBOI against arc-level simulation (50 000 scenarios) on 168 plans; z: difference over its simulation standard error.

Measure	Plans	Mean |err| (%)	Max |err| (%)	Share $\|z\| < 3$
Max	168	0.041	0.137	0.994
Range	168	0.062	0.249	1.000
GMD	168	0.057	0.241	1.000
MAD	168	0.064	0.272	0.988
Var	168	0.108	0.440	0.994

Table 3: Believed against true ex-post value (mean signed %, 50 runs per cell; truth: simulation without the day factor, which is what each model represents). Last column: largest absolute PBOI error for any plan.

	Believed vs true (mean signed %)				PBOI
Measure	Det	SAA	SORTIE-0	SORTIE-1	max |err| %
Max	-7.97	-0.71	-0.01	-0.01	0.20
Range	-17.56	-0.99	+0.03	+0.02	0.70
GMD	-14.68	-0.91	+0.04	+0.05	0.61
MAD	-16.17	-0.88	+0.05	+0.07	0.68
Var	-27.23	-1.33	+0.00	+0.02	0.98

and the PBOI value believed by SORTIE has a mean signed error below 0.1%. In line with Proposition 10, the cost-optimal plans, whose routes are far from balanced, are misjudged by only 4–6%.

Table 4 reports the out-of-sample ex-post value relative to Det on the 25 instances, five measures and two budgets. Pooled over the 250 runs, SORTIE-0 improves on Det by 0.89% on average (median 0.23%; better in 141 runs, tied in 96, worse in 13; $p < 10^{-18}$) and SORTIE-1 by 0.77% (better in 154, worse in 31). SAA improves by only 0.14% (better in 115, worse in 74), and both SORTIE variants beat it head to head ($p < 10^{-17}$). The largest gain is for the maximum at $B = 1.10$ (2.1%, 24 of 25 instances). The gains are modest because under homogeneous noise a route's standard deviation grows with its length, so balancing lengths roughly balances risks: the deterministic plan's *choice* is often close to the best one, while its *assessment* of its own fairness is badly wrong. For context, the cost-optimal plan has an expected realised range of 2087 against about 1340 for the fair plans, at a 6.5% higher cost.

Figure 2 shows what the drivers of a representative instance (X641-n30-b0, whose SORTIE-0 gain is close to the median) experience. The realised workloads are wide and right-skewed, and the expected realised range of both fair plans is about 1225, 30% above the deterministic range of about 940 that the Det planner reports.

Route consistency. Without the lower level, i.e. letting the planner sequence each route freely, 23 of 25 Det plans and 24 of 25 SORTIE plans for the range contain a route that is not a shortest tour of its customers, with an average excess length of 95 and 113 distance units. These detours lower the measured unfairness by 1–3%, and they are exactly the behaviour signalled by negative rank prices and excluded by the bilevel constraint (E4 in the Online Supplement).

Table 4: Improvement of the out-of-sample ex-post value over Det (mean %, positive is better; in brackets the number of instances, of 25, on which the method is strictly better). *: Wilcoxon signed-rank $p < 0.05$ in favour of the method.

Measure	B	SAA	SORTIE-0	SORTIE-1
GMD	1.05	-0.40 (6/25)	-0.09 (8/25)	-0.03 (9/25)
GMD	1.10	-0.32 (9/25)	+0.94* (14/25)	+0.34* (17/25)
MAD	1.05	+0.86* (11/25)	+1.10* (13/25)	+1.24* (12/25)
MAD	1.10	+0.70 (13/25)	+0.96* (14/25)	+1.53* (18/25)
Max	1.05	+0.71* (14/25)	+0.61* (15/25)	+0.77* (15/25)
Max	1.10	+1.70* (23/25)	+2.08* (24/25)	+1.93* (24/25)
Range	1.05	+0.12 (13/25)	+0.64* (15/25)	+1.56* (15/25)
Range	1.10	-0.21 (13/25)	+1.07* (17/25)	+0.39* (15/25)
Var	1.05	+0.48 (7/25)	+0.50* (9/25)	+0.48* (13/25)
Var	1.10	-2.25 (6/25)	+1.09* (12/25)	-0.49* (16/25)

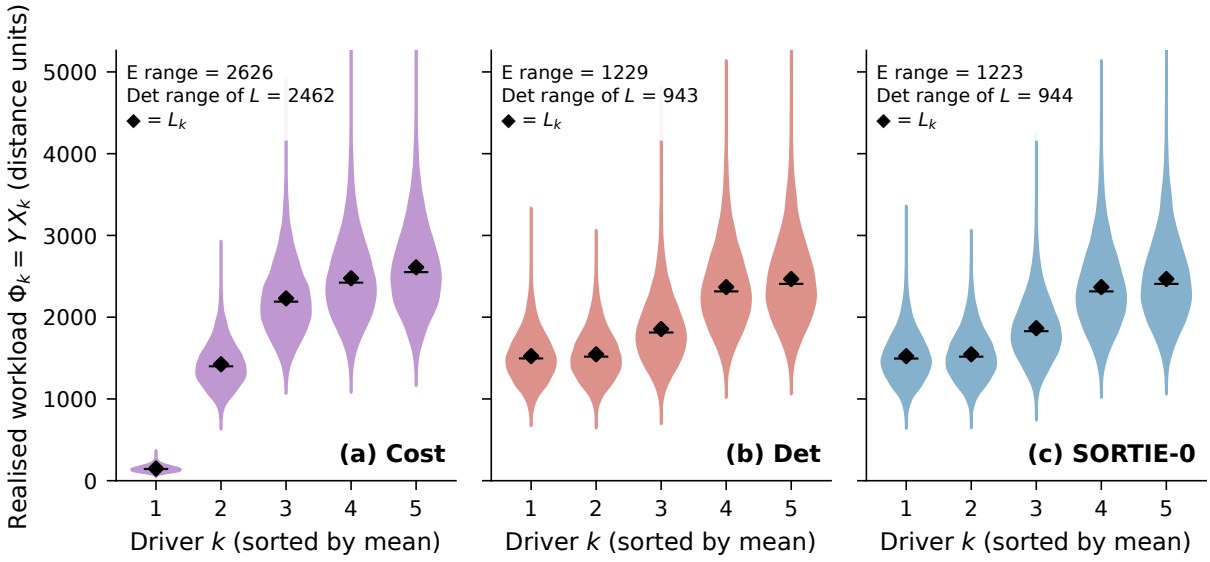


Figure 2: Simulated realised workloads $\Phi_k = Y X_k$ of the cost-optimal, Det and SORTIE-0 range plans of X641-n30-b0 (20 000 days); diamonds mark the tour lengths seen by a deterministic model.

6.4 The budget frontier

Figure 3 varies the budget factor from $B = 1.02$ to 1.30 on the ten instances with $n \in \{25, 35\}$. The gain of SORTIE-0 over Det grows with the budget, from about 1% for $B \leq 1.10$ to 3.2% at $B = 1.15$ and 6.8% at $B = 1.30$, where SORTIE-0 is better on all 30 instance–measure pairs. With little slack every fair plan is close to the cost-optimal one; with more slack many plans have about the same deterministic range but different realised ranges, and only the ex-post objective can tell them apart. For $B \geq 1.15$ SORTIE-0 also uses *less* of the budget: at $B = 1.30$ the Det plans for the maximum have a total length 20.5% above C^* and an expected realised maximum 5.2% below that of the cost plan, whereas the SORTIE-0 plans use 13.5% and achieve 8.7%. The deterministic objective spends extra length on routes whose deterministic length is irrelevant to the maximum but whose variance raises the realised one.

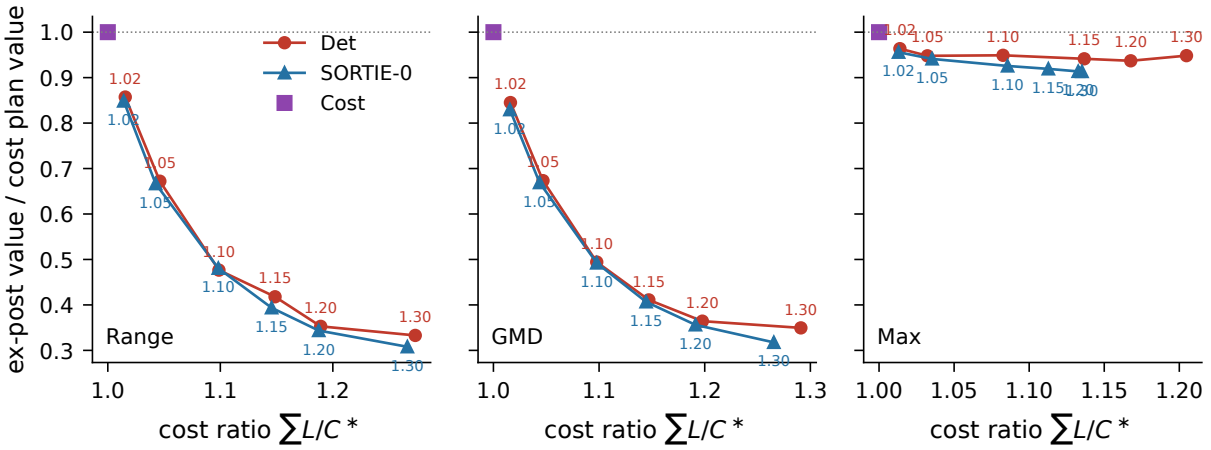


Figure 3: Price of fairness: mean total length relative to C^* against the mean expected realised unfairness relative to the cost-optimal plan, for budget factors B from 1.02 to 1.30 (labels), $n \in \{25, 35\}$.

Table 5: Simpler alternatives, $B = 1.10$: improvement over Det of the risk-adjusted model (Det-σ), of PBOI with normal marginals (PBOI-N) and of SORTIE-0 on the 25 main-grid instances (format as Table 4; the variance is omitted for PBOI-N, whose value does not depend on the marginal family for that measure).

Measure	Det-σ vs Det	PBOI-N vs Det	SORTIE-0 vs Det	SORTIE-0 vs Det-σ
Range	-0.49 (8/25)	+0.48* (18/25)	+1.07* (17/25)	+1.51* (19/25)
GMD	-0.69 (8/25)	+0.83* (14/25)	+0.94* (14/25)	+1.50* (18/25)
Max	-0.08 (8/25)	+2.08* (24/25)	+2.08* (24/25)	+2.13* (25/25)
Pooled	-0.81† (37/125)	+1.10* (71/100)	+1.23* (81/125)	+1.86* (97/125)

6.5 Simpler alternatives

Two natural alternatives to the exact integral are a larger sample and a risk-adjusted deterministic model (Table 5). SORTIE-0 is better than SAA with $S \in \{50, 200, 1000, 5000\}$ scenarios by 1.4, 1.6, 1.1 and 0.9%, significantly for $S \leq 1000$; at $S = 5000$ SAA is still worse on 9 pairs and better on 2 of 30, but no longer significantly ($p = 0.05$). Det-σ, which balances $L_k + \sigma_k$, is *worse* than Det (−0.8%, better on 37 and worse on 61 of 125 pairs, $p = 0.02$), and SORTIE-0 beats it by 1.9% (97 of 125 pairs). Adding a safety margin to every route treats each route's risk separately, whereas the expected ordered measure depends on how likely each route is to occupy each rank, which is what the Poisson–binomial integral and the rank prices capture. Replacing the gamma marginals of PBOI by normal ones changes nothing significant ($p = 0.65$).

6.6 A second benchmark family

All instances so far are slices of one file, and slicing places the depot at the first node of each block. We therefore solved the depot and the first 50 customers of twelve further CVRPLIB X instances (Uchoa et al., 2017), which were generated with different depot positions and customer distributions, with $m = 5$, the same capacity rule, $B = 1.10$, 120 s and two solver seeds (Table 6). The illusion reaches 40% for the range and 35% for the GMD, against 23% and 19% on the sliced instances at the same budget. SORTIE-0 improves on Det by 2.2% on average (median 1.7%;

Table 6: Second benchmark family (twelve X instances, depot and first 50 customers), $B = 1.10$, mean of two solver seeds: improvement of the ex-post value over Det (format as Table 4).

Measure	SAA vs Det	SORTIE-0 vs Det	Det-σ vs Det	Det belief err. (%)
Range	+1.48 (8/12)	+2.91* (11/12)	-0.39 (5/12)	-40.3
GMD	-0.62 (5/12)	+1.64 (8/12)	-2.15† (1/12)	-34.7
Max	+1.36* (10/12)	+1.93* (12/12)	+0.01 (7/12)	-8.4
Pooled	+0.74* (23/36)	+2.16* (31/36)	-0.84 (13/36)	-27.8

better on 31 and worse on 5 of 36 pairs, $p < 10^{-6}$), with a positive average on every one of the twelve instances (0.7 to 4.2%); SAA improves by 0.7%, and SORTIE-0 beats it on 33 of 36 pairs.

6.7 Robustness and scale

Noise model. Re-evaluating the plans under lognormal, incident-type (an arc takes 2.5 times as long with probability 0.02) and spatially correlated noise with the same per-arc mean and variance raises the mean absolute PBOI error from 0.16% to 0.29, 1.4 and 1.9% respectively, but SORTIE-0 remains significantly better than Det under every family (1.0 to 1.2%). *Congestion.* With a noise multiplier between 1/3 and 2 that is largest near a congestion centre, the illusion for the range grows from 23% to 26% and the gain of SORTIE-0 to 1.5% (53 of 75 pairs better, 3 worse). *Fleet size and noise level.* The gain grows with the number of vehicles (3.2% with seven vehicles) and with the noise level (2.1% for the range at $v = 0.5$). *Size.* At $n = 100$ single runs are dominated by search noise; averaging three solver seeds of 240 s, SORTIE-0 improves on Det by 4.2% (13 of 18 instance–measure pairs, $p = 0.02$), while the difference from SAA (1.1%) is not significant. Strikingly, the Det plans for the range vary by 11% across seeds, against 3% for SORTIE-0: the deterministic range is flat over many plans whose realised fairness differs. Details are in experiments E6–E8, E10 and E12 of the Online Supplement.

6.8 Managerial insights

(i) Report realised, not planned, fairness. A deterministic fair plan understates the unfairness its drivers will experience, by more the more balanced the plan; PBOI computes the realised value in under a millisecond for any plan, whatever method produced it. *(ii) Do not spend the budget on deterministic balance.* With generous budgets a deterministic fair model spends the slack on equalising route lengths, which does little for realised fairness; the ex-post objective achieves more with less. *(iii) Neither safety margins nor more scenarios are a substitute.* Balancing risk-adjusted lengths is worse than ignoring uncertainty, and a sample average needs thousands of scenarios to approach the exact objective. *(iv) Keep the shortest-tour requirement.* Without it, fairness is bought with detours; under uncertainty it must be imposed explicitly, and the risk-consistent variant changes few tours in practice (E13).

7 Conclusions

Fair routing under random travel times has to say whose fairness it measures. Drivers experience realised durations, so the relevant quantity is the expected unfairness of the realised day. A

deterministic fair plan cannot see this quantity: in our experiments it understated it by 8 to 27% on the sliced benchmark instances and by up to 40% on a second instance family, and Proposition 10 shows why the understatement is largest for the balanced plans that fairness optimisation produces. Fleet-wide shocks, by contrast, leave the expected value of every ordered measure unchanged, although they inflate the tail of unfairness.

The Poisson–binomial order-statistic integral evaluates any ordered measure of independent route durations exactly, in under a millisecond for five vehicles, and its derivatives, the rank prices, are both interpretable and useful: they identify the routes a fairness objective would like to lengthen, and they rank candidate moves with a Spearman correlation of about 0.9 with their exact effect. Under uncertainty route consistency no longer follows from monotonicity, so it has to be part of the model; the mean–standard-deviation lower level makes it computable by parametric Held–Karp.

The improvement of the plan itself is significant but modest at the standard budget, and it grows where the deterministic model has most freedom to choose wrongly: with more slack in the budget (up to 7%, with a lower total length), with more vehicles and on instances whose depot is not a block corner (2.2%, better on all twelve instances of the second family). It survives misspecified and spatially correlated noise, it is not reproduced by a risk-adjusted deterministic model, and a sample average needs thousands of scenarios to come close.

Limitations and future work. The lower level is exact only for routes of up to 15 customers; beyond that it relies on multi-start local search, and the bilevel constraint is then satisfied only approximately. PBOI assumes conditionally independent route durations; correlated congestion reduces its accuracy (to about 2% in our tests) although not the ranking of plans. At 100 customers single runs are noisy, the risk-consistent variant is too slow to converge, and SORTIE could not be separated statistically from a 200-scenario sample average. Exact methods for the stochastic bilevel problem (branch-and-cut with PBOI-based bounds), correlated route durations, multi-day fairness, in which unfairness can be compensated across days, and time-dependent travel times are natural next steps.

Declaration of competing interest

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CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Formal analysis, Software, Writing – original draft, Supervision. **Hoang-Viet Vu:** Software, Investigation, Validation, Visualization,

Writing – review & editing. **Phuc-Son Nguyen:** Formal analysis, Validation, Writing – review & editing.

Data availability

The instances (CVRPLIB X instances), the code, all solved plans, the raw results and the scripts that generate every table and figure will be made publicly available in a repository upon acceptance and are available from the corresponding author on request.

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